



Mindanao State University – Iligan Institute of Technology

Department of Electrical Engineering

ENS161: Statics of Rigid Bodies
B5-1

PROBLEM SET 6
PROBLEM SET 7
QUIZ 7

ZERO FORCE MEMBERS
B5-1

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Part I. Conceptual Questions

Instructions: For fill-in-the-blank questions, each blank is worth a point. For true or false questions, write only T or F. For identification questions, each desired item is worth a point.

1. The _____ of a rigid body represents it as having being isolated from its surroundings, replacing physical supports with forces and couple moments.

Answer: Free-Body Diagram (FBD)

Explanation: This is explained in 14th edition Engineering Mechanics Statics by R.C Hibbeler, Chapter 5.2 p.209.

2. **True or False.** An ideal truss does not have three-force members.

Answer: F

3. For 3D systems, the internal moment has two components: _____ and _____, whose axes are tangent and normal to the cross section, respectively.

Answer: Torsional Moment, Bending Moment

Note provided: Torsional moment (v_x , Shear Force), Bending Moment (M_x , Couple Moment).

4. A body is said to be _____ constrained if it experiences a fewer number of reactive forces than there are applicable equations of static equilibrium.

Answer: Partially

Explanation: This is explained in 14th edition Engineering Mechanics Statics by R.C Hibbeler, Chapter 5.2 p.253.

5. **True or False.** A beam is subject to a load uniformly distributed throughout its length. The beam is supported at its left end by a pin, at its right end by a rocker, and at its midpoint by a roller. The beam is statically determinate.

Answer: F

Explanation: A pin has 2 reactions, both rocker and roller have 1, in total of 4 reactions. The equations of equilibrium are 3. A structure is statically determinate if the number of unknown external reaction forces equals the number of available equations of static equilibrium. Thus $4 > 3$, and the system is **statically indeterminate**.

6. For a simply supported thirteen-member plane truss to be considered simple, it must have _____ joints.

Answer: F

Explanation: A ball-and-socket joint prevents any translation of the connecting member, but the connecting member is allowed to rotate freely about any axis; no couple moment is resisted. This is explained in 14th Edition Engineering Mechanics Statics by R.C Hibbeler, Chapter 5.7 p. 252.

7. **True or False.** A frictionless ball-and-socket joint resists translations and rotations.

Answer: F

Explanation: A ball-and-socket joint prevents any translation of the connecting member, but the connecting member is allowed to rotate freely about any axis; no couple moment is resisted. This is explained in 14th Edition Engineering Mechanics Statics by R.C Hibbeler, Chapter 5.7 p. 252.

8. A body is considered constrained if all the reactive forces intersect at a common point or pass through a common axis, or if all reactive forces are parallel.

Answer: Improperly

Explanation: In three dimensions, a body will be improperly constrained if the lines of action of all the reactive forces intersect a common axis. This is explained in 14th Edition Engineering Mechanics Statics by R.C Hibbeler, Chapter 5.7 p. 252.

9. **True or False.** A stable body requires that the lines of action of the reactive forces do not intersect a common axis and are not parallel to one another.

Answer: T

Explanation: Having the same number of unknown reactive forces as available equations of equilibrium doesn't always guarantee that a body will be stable when subjected to a particular loading. This is explained in 14th Edition Engineering Mechanics Statics by R.C Hibbeler, Chapter 5.7 p. 252

10. For internal loadings of 2D systems, the _____ force and the _____ force are normal and tangent to the cross section, respectively.

Answer: Normal, Shear

Explanation: Normal Force is the internal resultant force component that acts perpendicular (normal) to the cross section. The Shear Force is the internal resultant force component that acts parallel (tangent) to the cross section. This is explained in 14th Edition Engineering Mechanics Statics by R.C Hibbeler, Chapter 7.1 p. 344.

Scenario 1: Long Cantilever Beam

Problem Statement: A $1.1 - m$ long cantilever beam is subject to a $2 - kN$ weight at its free end.

1. True or False. This scenario is statically determinate?
2. The support reactions have magnitudes of ____ kN and ____ kN·m.
3. The maximum bending moment developed is ____ kN·m in magnitude and occurs ____ m from the free end (at the wall).
4. The maximum shear force developed is ____ kN in magnitude and occurs ____ m from the free end (constant).
5. The graph of the shear force as a function along the length of the beam is that of a polynomial of degree ____.
6. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree ____.

1. Reference: Simple Lever Force Diagram

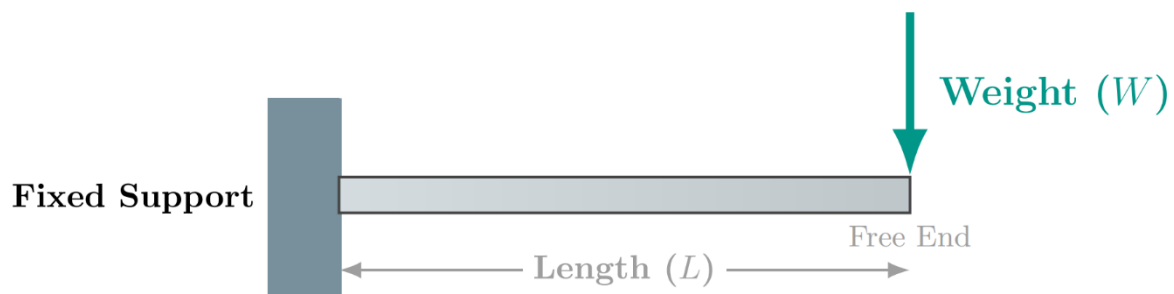


Figure 1: Conceptual Simple Lever Diagram

2. Scenario 1: Free Body Diagram & Determinacy

Support Type: Fixed Support (Wall).

Reactions: Vertical Force (A_y), Horizontal Force (A_x), Moment (M_A).

Determinacy: Unknowns (3) = Equilibrium Equations (3)
Therefore, the beam is **Statically Determinate**.

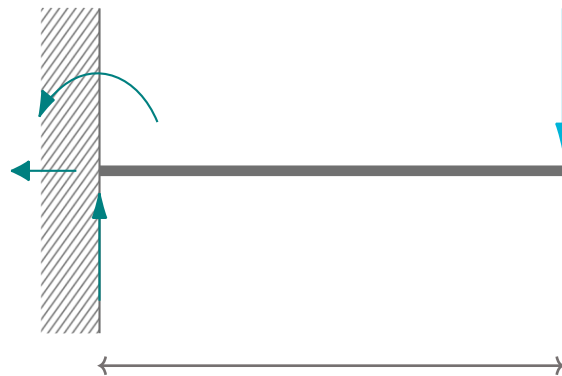


Figure 2: Cantilever FBD

3. Calculation of Reactions

Using the Equations of Equilibrium ($\sum F_x = 0$, $\sum F_y = 0$, and $\sum M = 0$):

A. Sum of Vertical Forces:

$$\begin{aligned}\sum F_y &= 0 \\ A_y - W &= 0 \\ A_y &= 2 \text{ kN } (\uparrow)\end{aligned}$$

B. Sum of Moments at Support (A):

$$\begin{aligned}\sum M_A &= 0 \\ M - (WL) &= 0 \\ M_A &= 2 \text{ kN } (1.1 \text{ m}) \\ M_A &= 2.200 \text{ kN} \cdot \text{m } (\curvearrowright)\end{aligned}$$

4. Calculation of Shear and Moment

Let x be the distance from the **free end** (moving Right to Left), where $0 \leq x \leq 1.1 \text{ m}$.

Shear Force (v): We sum the vertical forces

$$\begin{aligned}\sum F_y &= 0 \\ v - (2 \text{ kN}) &= 0 \\ v &= 2 \text{ kN}\end{aligned}$$

Since there is no variable, the degree is 0 (constant).

Bending Moment (M): Sum the moments

$$\begin{aligned}\sum M &= 0 \\ M + M_A - (2 \cdot x) &= 0 \\ M + 2.2 - 2x &= 0 \\ M(x) &= 2x - 2.2 \text{ kN} \cdot \text{m}\end{aligned}$$

Since the variable is x , the degree is 1 (Linear).

Verification at boundaries:

- At Free End ($x = 0$):

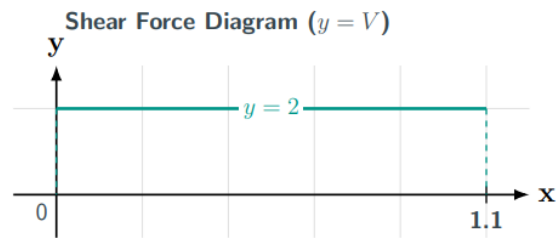
$$M(0) = -2(0) = 0 \quad [\text{Correct}]$$

- At Fixed Support ($x = 1.1$):

$$M(1.1) = 2(1.1) - 2.2 = 0 \text{ kN} \cdot \text{m} \quad [\text{Correct}]$$

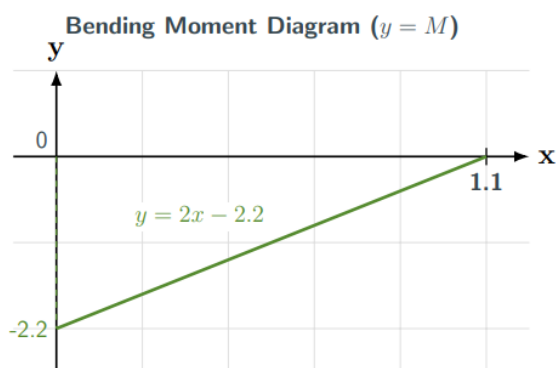
5. Shear and Moment Diagram:

A. Shear Force Diagram: The shear force is the internal vertical force resisting the external load. Starting from the free end, the only vertical force acting on the beam is the 2 kN downward load. To maintain equilibrium, every cross-section along the beam must provide an equal and opposite upward force of 2 kN . Since there are no other loads between the free end and the fixed support, this value remains constant at 2 kN throughout the entire length ($L = 1.1\text{ m}$).



B. Moment Diagram: The bending moment represents the internal torque trying to bend the beam. At the free end ($x = 1.1$), the moment is zero because there is no distance from the load. As we move closer to the fixed support ($x = 0$), the distance from the load increases. Since $\text{Moment} = \text{Force} \times \text{Distance}$, the magnitude of the moment increases linearly.

- At the **Fixed Support** ($x = 0$), the moment is maximum: $M(0) = -2.2\text{ kN} \cdot \text{m}$.



6. Answer

Based on the calculations and diagrams above:

1. **True** or False. This scenario is statistically determinate? **T**
2. The support reactions have magnitudes of **2** kN and **2.2** kN·m.
3. The maximum bending moment developed is **2.2** kN·m in magnitude and occurs **1.1** m from the free end (at the wall).
4. The maximum shear force developed is **2** kN in magnitude and occurs **0 to 1.1** m from the free end (constant).
5. The graph of the shear force as a function along the length of the beam is that of a polynomial of degree **0**.
6. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree **1**.

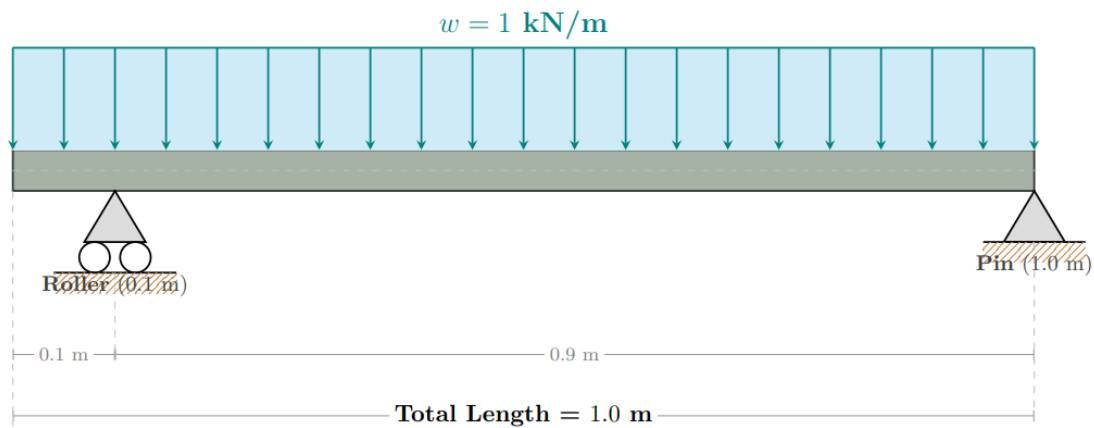
Scenario 2: Long Simple Beam

Problem Statement: A 1-m long beam carries a load uniformly distributed throughout its length. Assume a roller support 0.1 m from the left end and a pin support at the right end. The load has an intensity of 1 kN/m .

1. The roller support develops a force of kN.
2. The pin support develops a force of kN.
3. The bending moment developed at the right end is kN·m.
4. The shear force developed just to the left of the pin is kN.
5. The shear force developed just to the right of the pin is kN.
6. True or False. There is zero shear force developed at a point 0.9 from the right end.

7. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree ____.
8. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree ____.

1. Free Body Diagram



2. Support Reactions

Total resultant of UDL:

$$W = w \times L = 1.00 \times 1.00$$

$$W = 1.00 \text{ kN (downward)}$$

Vertical equilibrium:

$$\sum F_y = 0$$

$$R_{RY} + R_{PY} - W = 0$$

$$R_{RY} + R_{PY} = 1.00 \text{ kN}$$

The moment about the pin at $(x = 1.00 \text{ m})$ (CCW positive) is found by noting that the UDL acts at $(x = 0.50 \text{ m})$, with a moment arm of (0.50 m) , while the roller at $(x = 0.10 \text{ m})$ has a moment arm of (0.90 m) .

$$\sum M = 0$$

$$R_{RY} \times 0.90 - 1.00 \times 0.50 = 0$$

$$R_{RY} = 0.556 \text{ kN}$$

$$R_{PY} = 1.00 - 0.556 = 0.444 \text{ kN}$$

Pin horizontal reaction:

$$R_{PX} = 0 \text{ (no horizontal external loads, still show on FBD)}$$

3. Internal Shear $V(x)$ and Bending Moment $M(x)$

Region I: $0 \leq x < 0.10$ (left of roller)

Vertical equilibrium:

$$-W \times x - V(x) = 0$$

$$V(x) = -w \times x$$

$$V(x) = -1.00 \times x \text{ kN}$$

Bending moment (integrate $V(x)$, $M(0) = 0$):

$$\sum M = 0$$

$$M(x) = \int V(x) \, dx$$

$$M(x) = -0.5 (x^2) \text{ kNm}$$

Region II: $0.10 \leq x \leq 1.00$ (between roller and pin)

Left segment now includes R_{RY} @ $x = 0.10$;

$$V(x) = R_{RY} - w \times x$$

$$V(x) = 0.556 - x \text{ kN}$$

Bending moment $\frac{dM}{dx} = V(x)$

$$M(x) = \int (R_{RY} - x) dx$$

$$M(x) = R_{RY}x - 0.5x^2 + C$$

Determine C using continuity at $x = 0.10$:

$$M(0.10^-) = -0.5(0.10)^2$$

$$M(0.10^-) = -0.005 \text{ kNm}$$

$$M(0.10^+) = 0.10R_{RY} - 0.5(0.10)^2 + C = -0.005$$

$$C = -0.10R_{RY}$$

Final bending moment:

$$M(x) = R_{RY}(x - 0.10) - 0.5x^2$$

$$M(x) = 0.556(x - 0.10) - 0.5x^2 \text{ kNm}$$

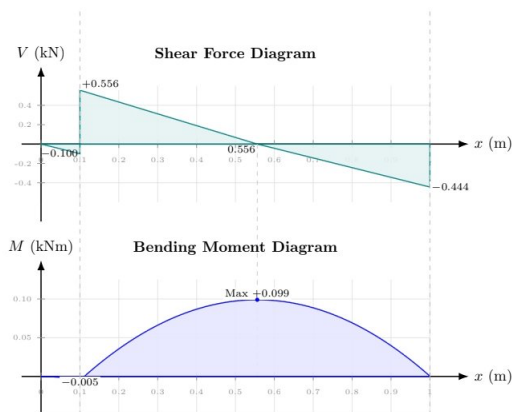


Figure 1: Complete Beam Analysis on Cartesian Planes for the Long Simple Beam.

4. Shear Just Left/Right of Pin & Max Moment

Shear just left of pin ($x = 1.00^-$):

$$V(1.00^-) = 0.556 - 1.00$$

$$V(1.00^-) = -0.444 \text{ kN}$$

Shear just right of pin ($x = 1.00^+$):

$$V(1.00^+) = -0.444 + 0.444$$

$$V(1.00^+) = 0 \text{ kN}$$

Maximum moment (where $V(x) = 0$):

$$V(x) = 0$$

$$0 = 0.556 - x$$

$$x = 0.556 \text{ m}$$

$$M_{max} = R_{RY}(x - 0.10) - 0.5x^2$$

$$M_{max} = 0.099 \text{ kNm}$$

6. Answers

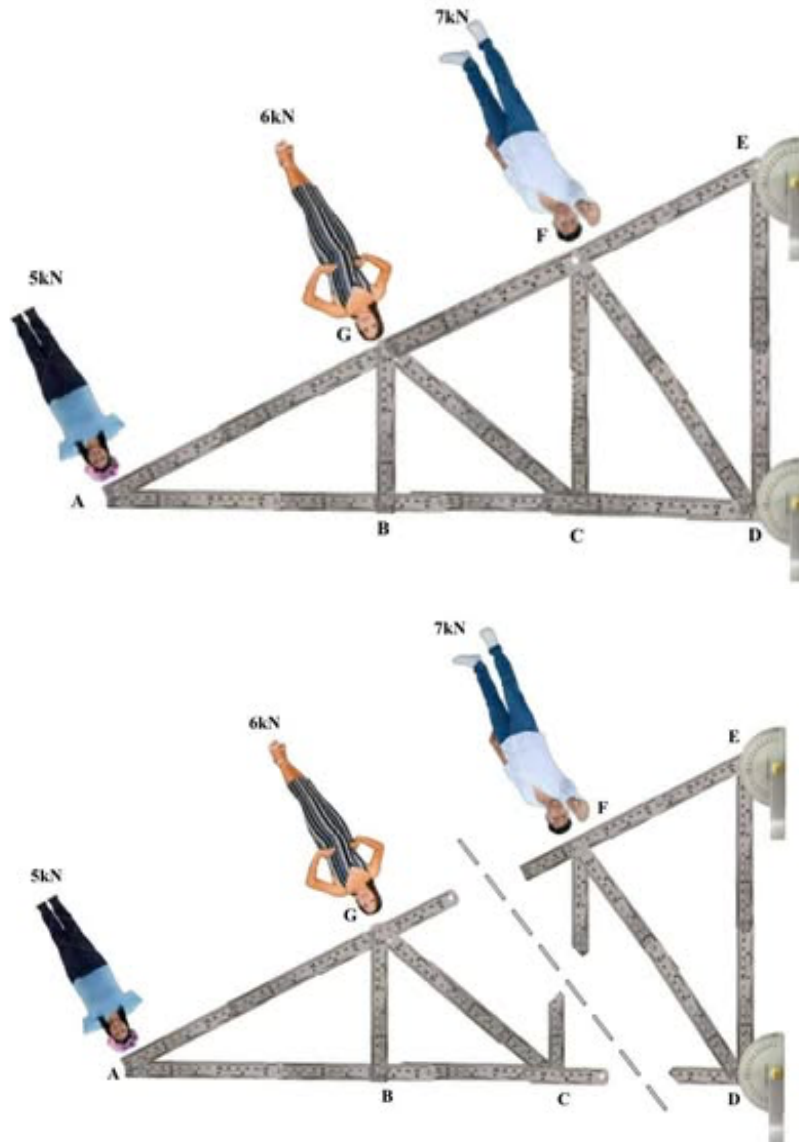
1. The roller support develops a force of **0.556** kN.
2. The pin support develops a force of **0.444** kN.
3. The bending moment developed at the right end is **0** kN-m.
4. The shear force developed just to the left of the pin is **-0.444** kN.
5. The shear force developed just to the right of the pin is **0.10** kN.
6. True or False. There is zero shear force developed at a point 0.9 from the right end. **F**
7. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree **1**.
8. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree **2**.

Part II. Problem Solving

Instructions: Where applicable, provide force diagrams and indicate the sense of member forces. Enclose final answers in boxes, with non-integer values rounded to three (3) decimal places.

Problem 1: Trusses

Problem 1. Consider the truss subject to idealized loads as shown above. Assume that the members have negligible weights and that the joints behave as frictionless pins. The magnitudes of p_1 , p_2 , and p_3 are 5 kN, 6 kN, and 7 kN, respectively. Set the dimensional parameters d and h to 2 ft and 3 ft, respectively.



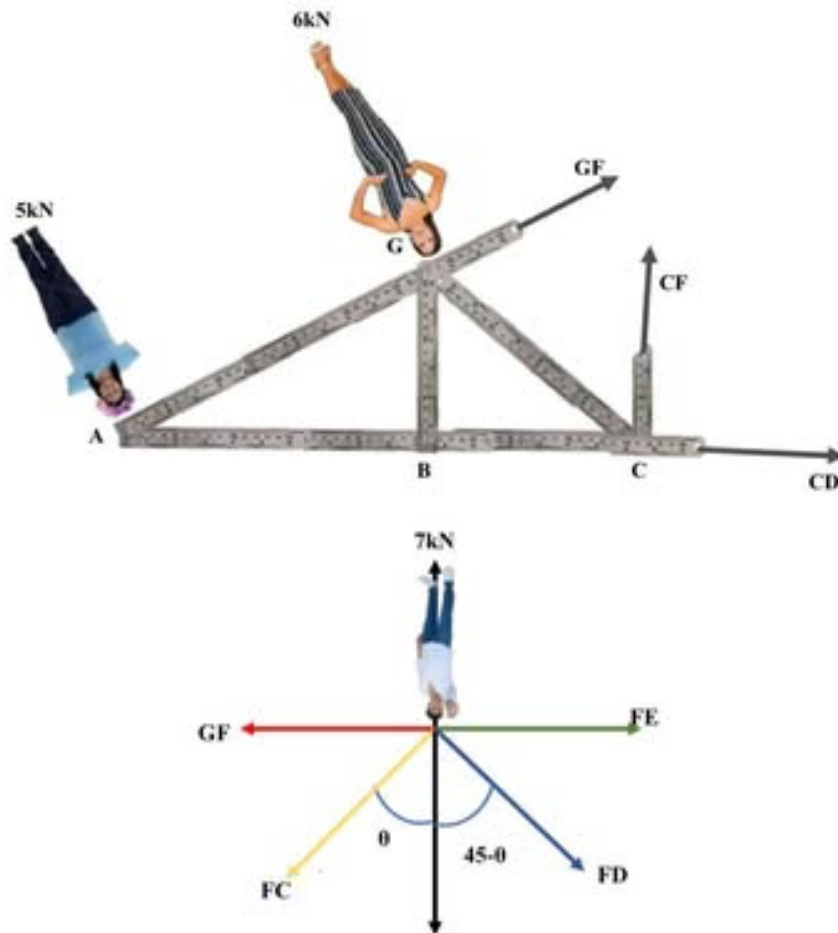
$$\theta = \tan^{-1}\left(\frac{3}{6}\right) = 26.56505^\circ$$

$$h_1 = \frac{3}{6}(4) = 2 \text{ ft}$$

$$h_2 = \frac{3}{6}(2) = 1 \text{ ft}$$

$$\phi = \tan^{-1}\left(\frac{2}{2}\right) = 45^\circ$$

a. (15points) Determine the forces developed in members FG, CF, and CD. Use the method of sections, making sure to indicate the section cut and show the corresponding free-body diagram.



Take the moment about F $\sum M_F = 0$, to get the F_{CD}

$$F_{CD}(2) + 6(\sqrt{5}) + 2(5)(\sqrt{5}) = 0$$

$$F_{CD} = \frac{-(6(\sqrt{5}) + 2(5)(\sqrt{5}))}{2}$$

$$F_{CD} = -17.8885 \text{ kN} \quad \text{or} \quad F_{CD} = 17.8885 \text{ kN (C)}$$

Take the horizontal resultant force of member F $\sum F_x = 0$ to get the F_{GF}

$$F_{GF} \cos(26.565^\circ) - 17.889 + 6 \sin(26.565) + 5 \sin(26.565) = 0$$

$$F_{GF} = \frac{17.889 - 4.919}{0.8944}$$

$$F_{GF} = 14.501 \text{ kN (T)}$$

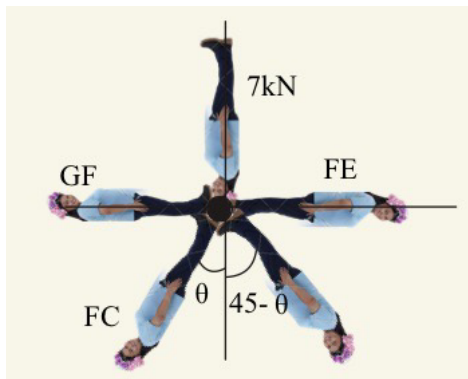
Take the vertical resultant force of member F $\sum F_y = 0$ to get the F_{CF}

$$14.501 \sin(26.565) + F_{CF} - 6 \cos(26.565) - 5 \cos(26.565) = 0$$

$$F_{CF} = 6 \cos(26.565) + 5 \cos(26.565) - 14.501 \sin(26.565)$$

$$F_{CF} = 3.354 \text{ kN (T)}$$

b. (10 points) Determine the forces developed in members DF and EF. Use the method of joints, making sure to show free-body diagrams at the joints considered. Hint: Use the computed forces in members FG, CF, and CD.



$$\sum F_Y = 0$$

$$7 \text{ kN} + F_{CF} \cos \theta + F_{DF} \cos(18.43495) = 0$$

$$F_{DF} = \frac{-7 - 3.354 \cos(26.56505)}{\cos(18.43495)}$$

$$F_{DF} = -10.541 \text{ kN} \quad \text{or} \quad \mathbf{F_{DF} = 10.541 \text{ kN (C)}}$$

$$\sum F_x = 0$$

$$F_{EF} + F_{GF} + F_{CF} \sin \theta + F_{DF} \sin(18.43495) = 0$$

$$F_{EF} = -(14.501 + 3.354 \sin \theta - 10.541 \sin(18.43495))$$

$$F_{EF} = -12.668 \text{ kN} \quad \text{or} \quad \mathbf{F_{EF} = 12.668 \text{ kN (C)}}$$

c. (5 points) If the truss is to be idealized as simply supported by a pin and a roller, at which joints must the pin and the roller be placed to maintain stability of the loaded truss? Justify your answer.

To maintain stability, the pin must be placed at joint A and the roller at joint D. Joint A must be the pin because the truss requires both a horizontal and vertical reaction to prevent sliding, and only a pin can provide two reaction components. Joint D should be the roller to provide a single vertical reaction and avoid over-constraining the structure. This pin-roller placement provides the required three reaction components (A_x , A_y , D_y) to keep the truss stable and statically determinate under the applied loads.

$$\sum F_Y = 0$$

$$A_y + D_y = P_1 + P_2 + P_3$$

$$A_y + D_y = 18 \text{ kN}$$

$$\sum M_A = 0$$

$$-(6 \times 2) - (7 \times 4) + D_y(6) = 0$$

$$\mathbf{D_y = 6.667 \text{ kN}}$$

$$\mathbf{A_y = 11.333 \text{ kN}}$$

$$\mathbf{A_x = 0 \text{ kN}}$$

d. (4 points) A contingency is considered wherein joint A is effectively unloaded. Identify zero-force members, if there be any, for this case.

When joint A is considered effectively unloaded, no external force or support reaction acts at that joint. The only members connected to A are AB and AC. According to the zero-force member rule for trusses, whenever two non-collinear members meet at a joint with no applied load or reaction, those members must carry zero force to satisfy equilibrium. In this case, since joint A has no loading and the attached members are not collinear, member AC becomes a zero-force member under this contingency. Thus, AC carries no internal force when joint A is unloaded.

e. (6 points) Determine the forces developed in the supports at E and at D. Hint: Use the computed forces in members CD, DF, and EF.

$$\sum F_x = 0$$

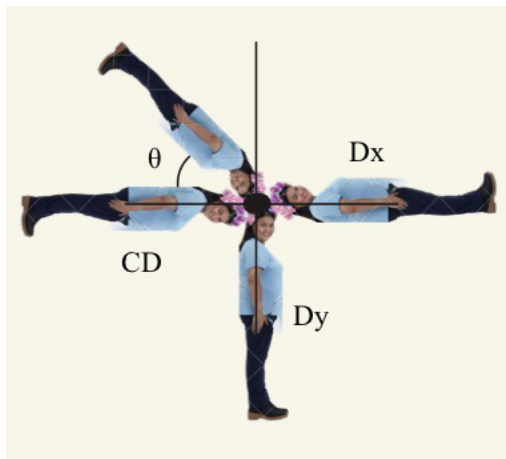
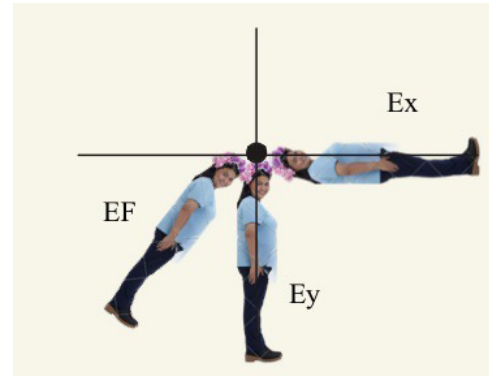
$$E_x = 12.668 \sin(63.43495)$$

$$E_x = \mathbf{11.331 \text{ kN} \rightarrow}$$

$$\sum F_y = 0$$

$$E_y = 12.668 \cos(63.43495)$$

$$E_y = \mathbf{5.665 \text{ kN} \downarrow}$$



$$\sum F_x = 0$$

$$D_x = -17.8885 - 10.541 \cos 45$$

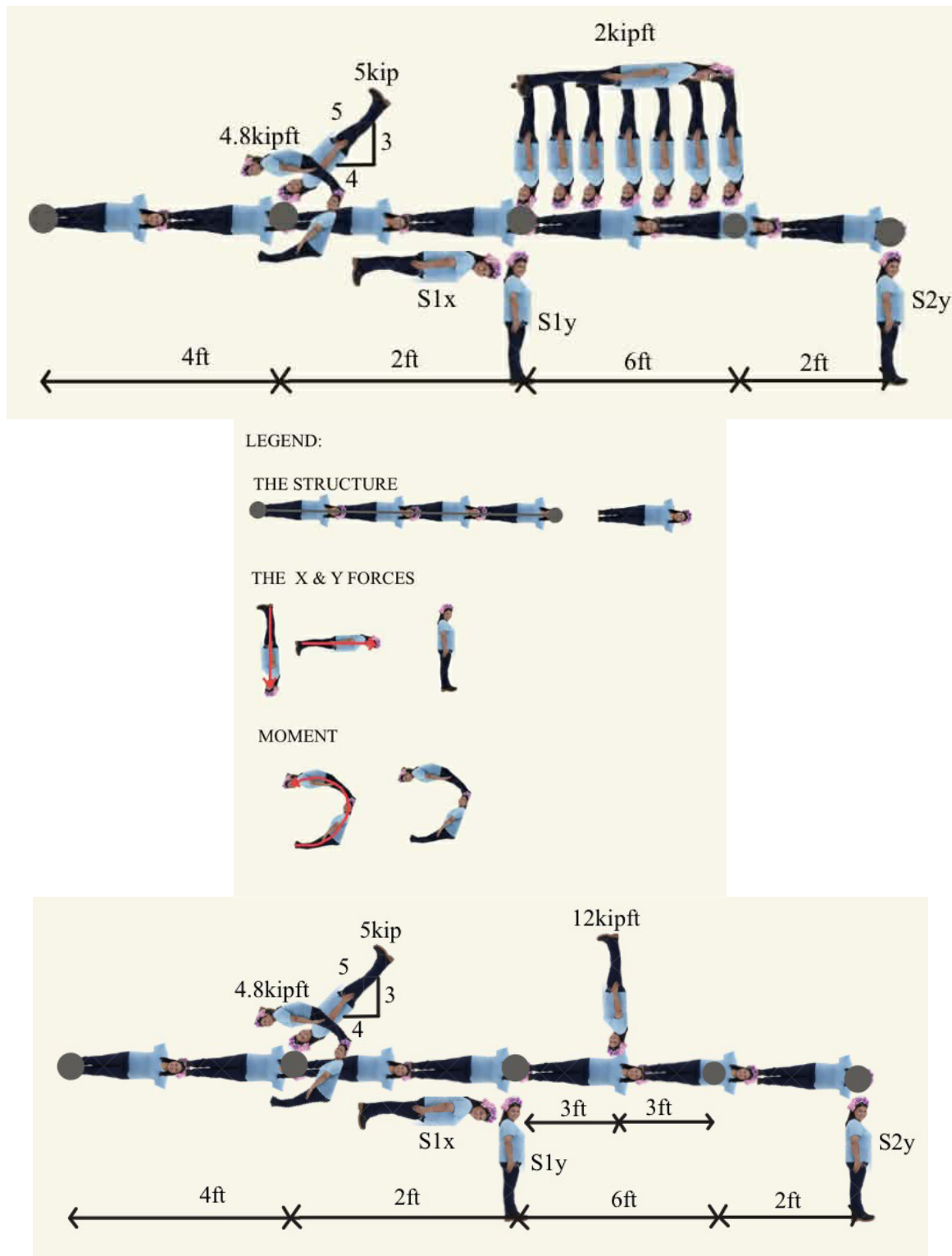
$$D_x = \mathbf{25.342 \text{ kN} \rightarrow}$$

$$\sum F_y = 0$$

$$D_y = -10.541 \sin 45$$

$$D_y = \mathbf{7.454 \text{ kN} \downarrow}$$

Problem 2: Shear Forces



Problem 2. A custom beam design places supports at S_1 and at S_2 , in anticipation of a concentrated load of u_{point} kips and a uniformly distributed load of intensity u_{dist} kip/ft. Your main task is to analyze how the internal loadings induced vary as a function of x , where x is the distance from the left end of the beam. Consider the case when u_{point} is 5 kips and u_{dist} is 2 kip/ft. Set the respective values of $d_1, d_2, d_3, d_4, d_5, d_6$ to 3.2, 0.8, 2.0, 6.0, 2.0, and 0.6 ft, and those of m_v and m_h to 3 and 4. For static determinacy, assume a pin at S_1 and a roller at S_2 .

- a) (6 points) Determine the support reactions.

Let point at (3.2, 0.6) be point A:

Solving for the **moment produced by u_{point}** about point A ft:

$$M_A = 5 \left(\frac{3}{5} \right) (0.8) + 5 \left(\frac{4}{5} \right) (0.6)$$

$$\mathbf{M_A = 4.8 \text{ kip} * ft}$$

Take moment about S_1 to solve for S_2 :

$$\sum M_{s_1} = 5 \left(\frac{3}{2} \right) (2) + 4.8 + S_2 (0.8) = 2(6) \left(\frac{6}{2} \right)$$

$$S_2 = \frac{36 - 4.8 - 6}{0.8}$$

$$\mathbf{S_2 = 3.15 \text{ kip}}$$

To find the external reactions of S_1 , using $\sum F_x = 0$ to find S_{1x}

$$S_{1x} = 5 \left(\frac{4}{5} \right)$$

$$\mathbf{S_{1x} = 4 \text{ kip}}$$

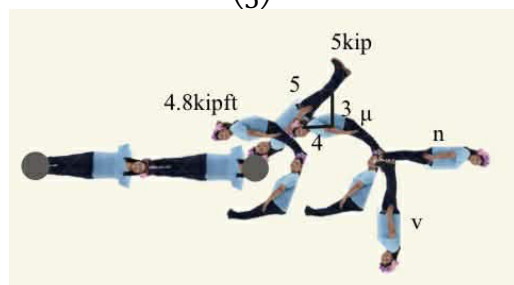
To find the external reactions of S_1 , using $\sum F_y = 0$ to find S_{1y}

$$S_{1y} = 5 \left(\frac{4}{5} \right) + 2(6) - 3.15$$

$$\mathbf{S_{1y} = 11.85 \text{ kip}}$$

- b) (1 point) Determine the normal force at $d_1 + d_2$ from the left end.

$$u = 5 \left(\frac{4}{5} \right) = \mathbf{4 \text{ kip}}$$



- c) (2 points) Determine the shear force at $d_1 + d_2$ from the left end.

$$v + 5 \left(\frac{4}{5} \right) = 0$$

$$\mathbf{v = -3 \text{ kip}}$$

- d) (3 points) Determine the bending moment at $d_1 + d_2$ from the left end.

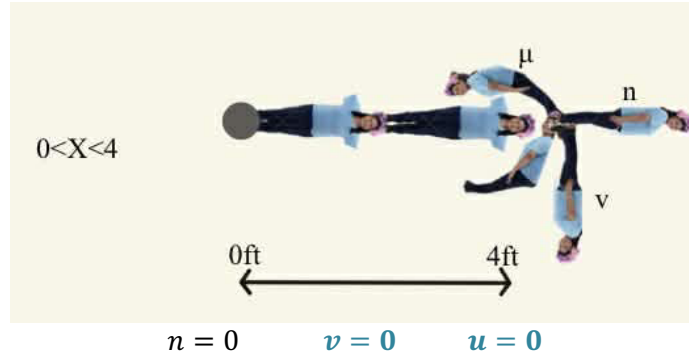
$$4.8 + u = 0$$

$$\mathbf{u = -4.8 \text{ kip} * ft}$$

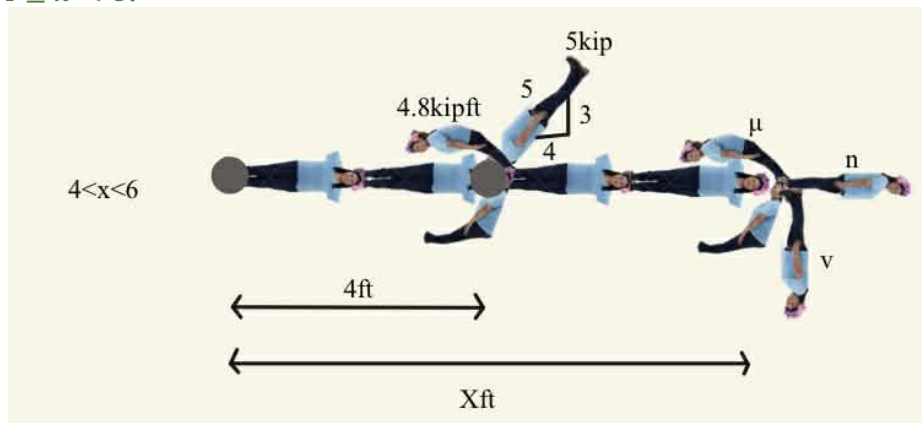
- e) (8 points) Determine the shear force as a function of x.

- f) (8 points) Determine the bending moment as a function of x.

Region @ $0 \leq x < 4$:



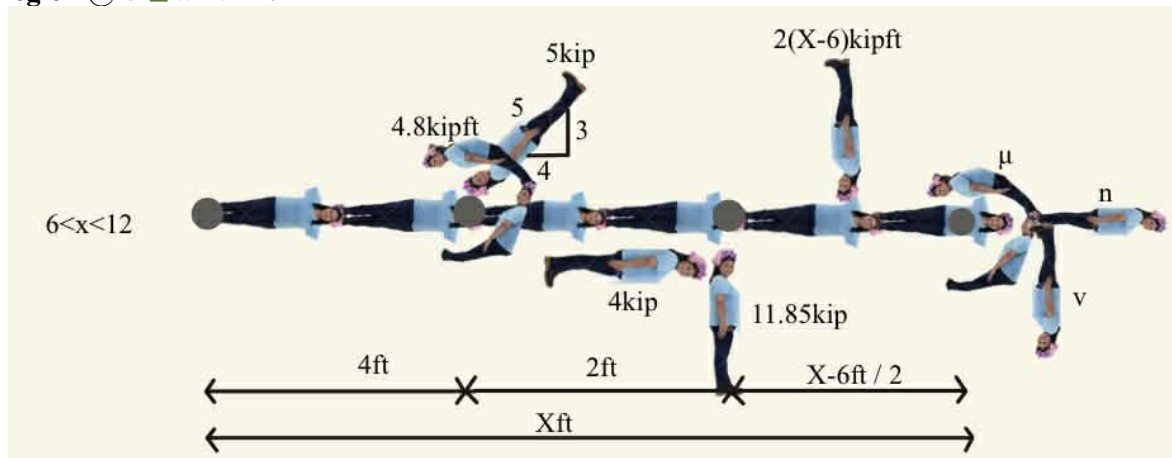
Region @ $4 \leq x < 6$:



$$u = -4.8 - 3x + 12$$

$$u = 7.2 - 3x$$

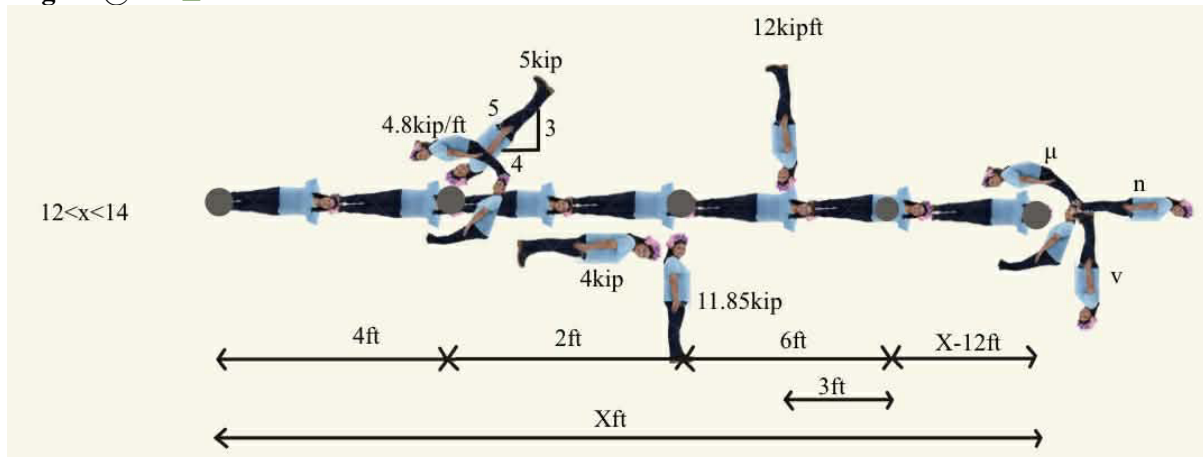
Region @ $6 \leq x < 12$:



$$u = -4.8 - 3x + 12 - (x^2 - 12x + 36) + 11.85x - 71.1$$

$$u = x^2 + 20.85x - 99.9$$

Region @ $12 \leq x < 14$:



$$\begin{aligned}
 n &= 0 \\
 v &= -5\left(\frac{3}{5}\right) - 2(6) + 11.85 \\
 v &= -3.15 \text{ kip} \\
 u &= -4.8 - 5\left(\frac{3}{5}\right)(x-4) - 2(6)(x-9) + 11.85(x-6) \\
 &= -4.8 - 3x + 12 - 12x + 108 + 11.85x - 71.1 \\
 u &= 44.1 - 3.15x
 \end{aligned}$$

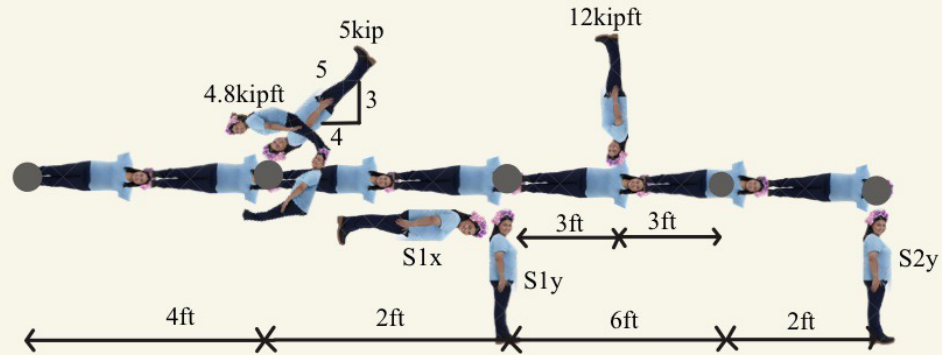
Shear Forces as a function of x:

$$\begin{aligned}
 v &= 0 & @ 0 \leq x < 4: \\
 v &= -3 \text{ kip} & @ 4 \leq x < 6: \\
 v &= 20.85 - 2x \text{ kip} & @ 6 \leq x < 12: \\
 v &= -3.15 \text{ kip} & @ 12 \leq x < 14:
 \end{aligned}$$

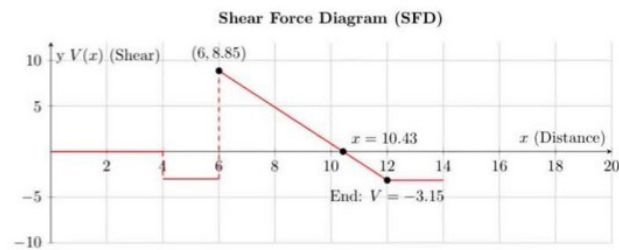
Bending Moments as a function of x:

$$\begin{aligned}
 u &= 0 & @ 0 \leq x < 4: \\
 u &= 7.2 - 3x & @ 4 \leq x < 6: \\
 u &= x^2 + 20.85x - 99.9 & @ 6 \leq x < 12: \\
 u &= 44.1 - 3.15x & @ 12 \leq x < 14:
 \end{aligned}$$

- g) (6 points) Plot the shear force diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations.



Beam Analysis Diagrams (Labeled)



- h) (6 points) Plot the bending moment diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations

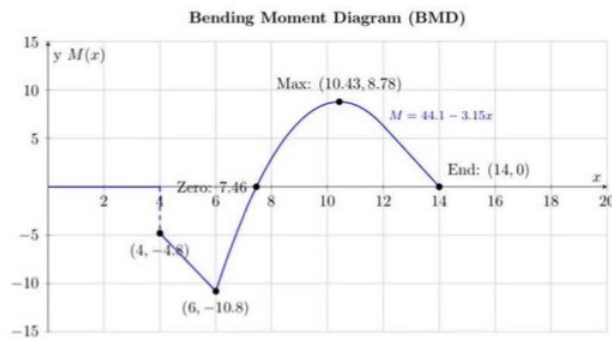
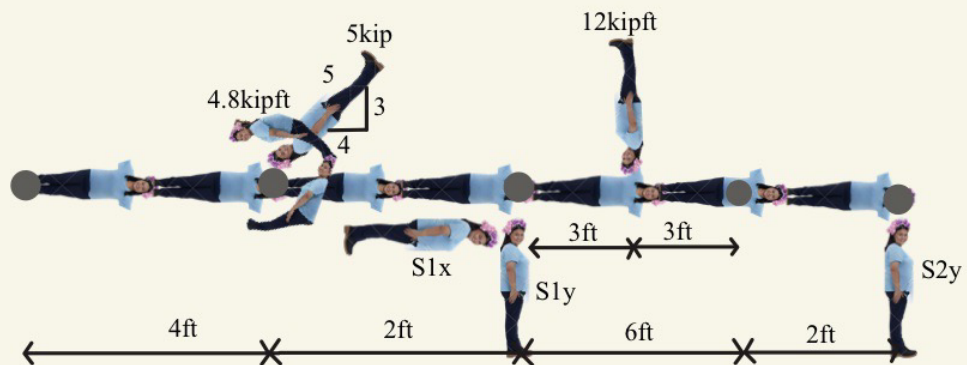


Figure 1: Beam Analysis Diagrams. The final section $12 \leq x < 14$ is now labeled with its equation and endpoint.